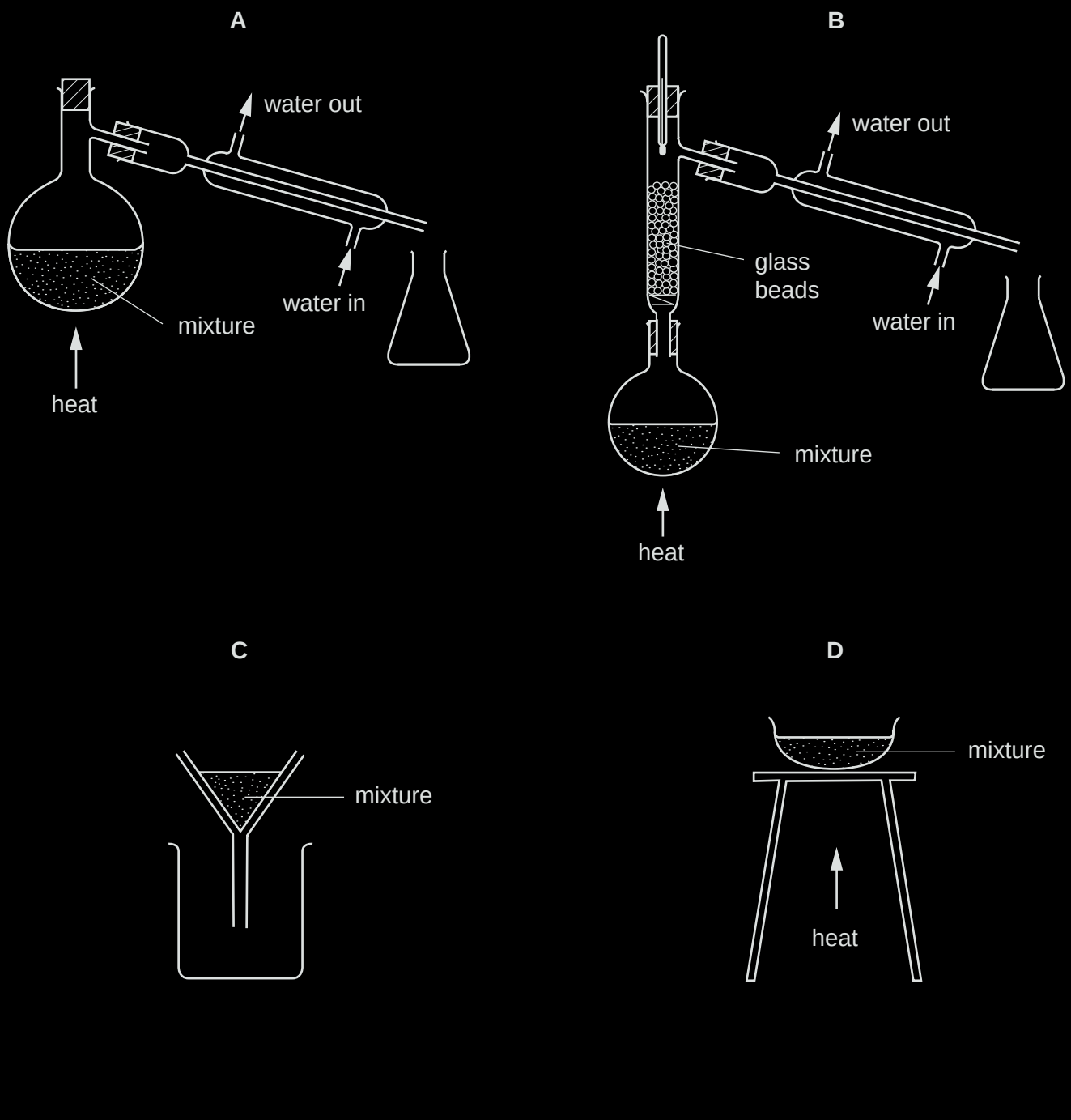
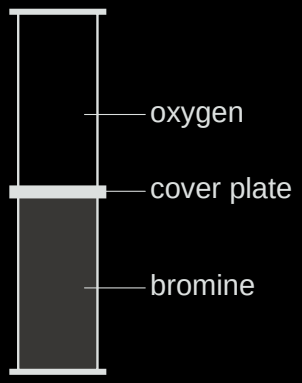

A
water out
water in
mixture
heat
B
water out
glass
beads
water in
mixture
heat
C
mixture
D
mixture
heat



Which statement explains this change?

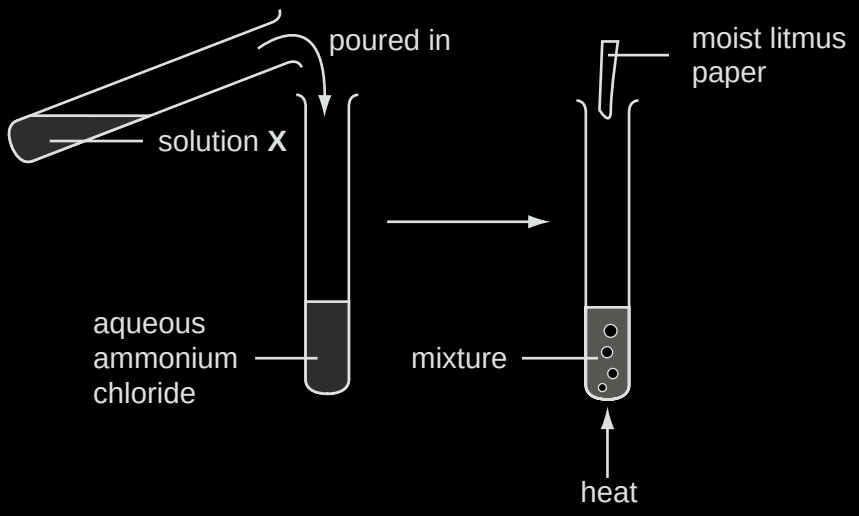
A Oxygen and bromine gases have equal densities.

B Oxygen and bromine molecules are in random motion.

C Oxygen and bromine molecules diffuse at the same rate.

D Equal volumes of oxygen and bromine contain equal numbers of molecules.

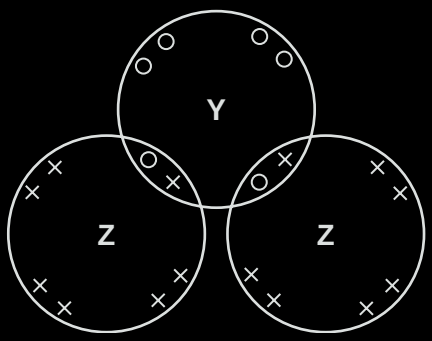
4 The diagrams show an experiment with aqueous ammonium chloride.



A gas, **Y**, is produced and the litmus paper changes colour.

What are solution **X** and gas **Y**?

	solution **X**	gas **Y**
A	aqueous sodium hydroxide	ammonia
B	aqueous sodium hydroxide	chlorine
C	dilute sulphuric acid	ammonia
D	dilute sulphuric acid	chlorine



key

○ outer electron of a **Y** atom

× outer electron of a **Z** atom

What are elements **Y** and **Z**?

	Y	**Z**
A	calcium	chlorine
B	carbon	oxygen
C	oxygen	hydrogen
D	sulphur	chlorine

8 Which **two** statements about a covalent bond are correct?

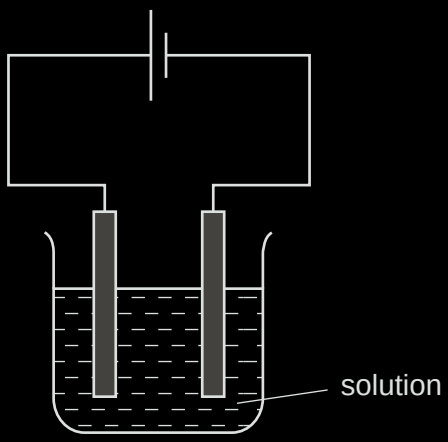
1 It can be formed between two metal atoms.

2 It can be formed between two non-metal atoms.

3 It is formed by the transfer of electrons between atoms.

4 It is formed by sharing electrons between atoms.

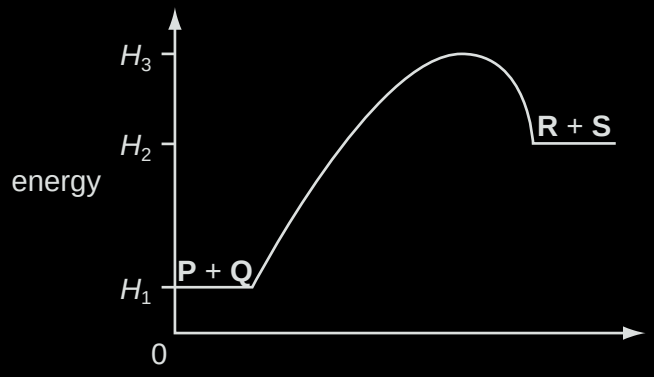
A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4



Which metal is deposited at the negative electrode and why?

	metal deposited	reason
A	copper	copper is less reactive than sodium
B	copper	copper is more reactive than hydrogen
C	sodium	copper is less reactive than hydrogen
D	sodium	copper is more reactive than sodium

13 The energy profile diagram below is for a reaction **P** + **Q** → **R** + **S**.



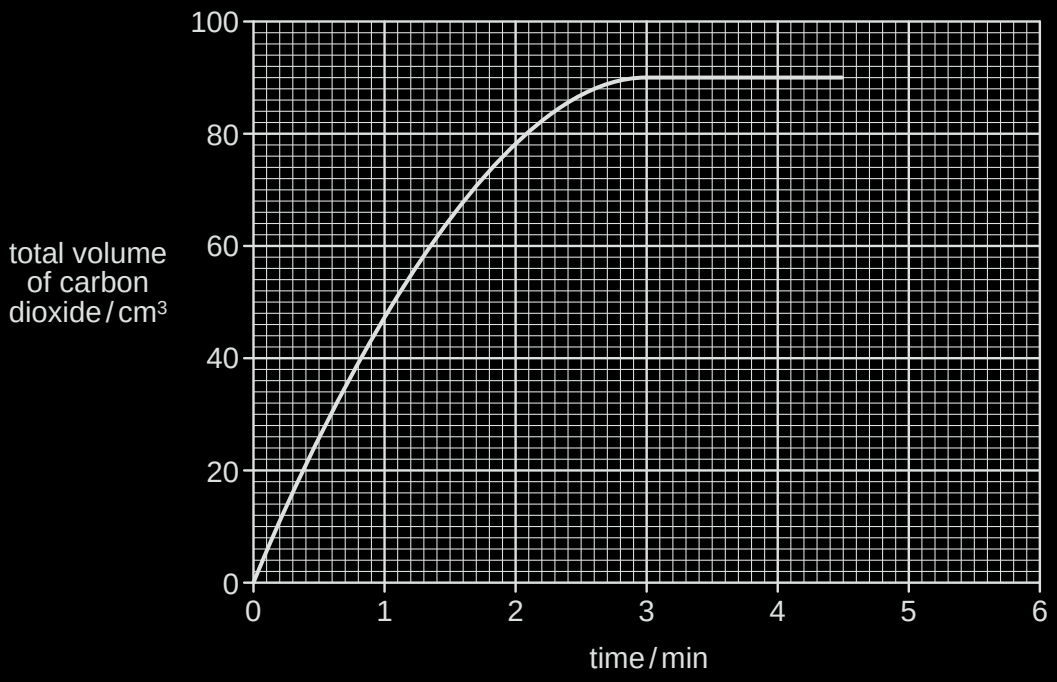
Which statement is correct?

A The activation energy of the reaction is ($H_3 - H_1$).

B The activation energy of the reaction is ($H_3 - H_2$).

C ΔH is ($H_1 - H_2$).

D ΔH is ($H_1 - H_3$).



How much time is required for half the calcium carbonate to react?

A 0.95 min **B** 1.5 min **C** 2.0 min **D** 3.0 min

15 Ammonia is made by a reversible reaction between nitrogen and hydrogen.

The equation for the reaction is shown.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \qquad \Delta H \text{ is negative}$$

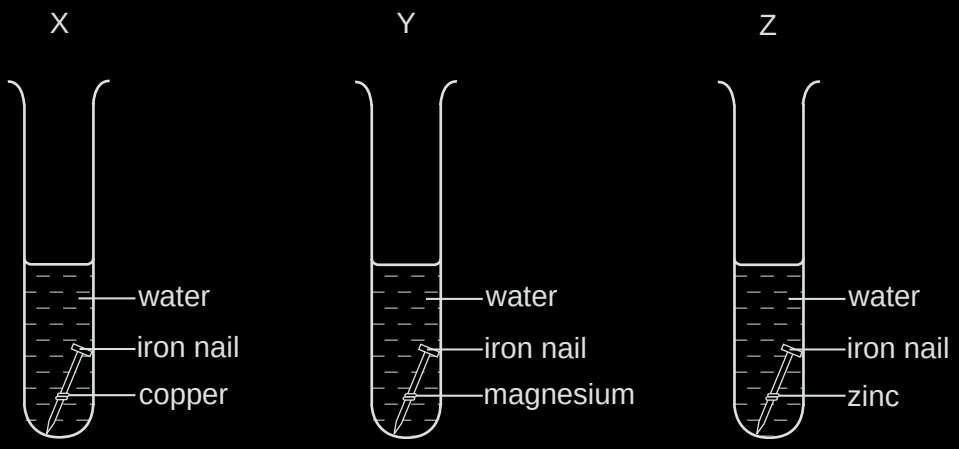
What is the effect of increasing the pressure in this process?

A Less ammonia is formed.

B Less heat is produced.

C More ammonia is formed.

D The reaction slows down.



In which test-tubes will the iron rust?

A X only **B** Y only **C** X and Z only **D** Y and Z only

28 One mole of compound **X** gives three moles of ions in aqueous solution. **X** reacts with ammonium carbonate to give an acidic gas.

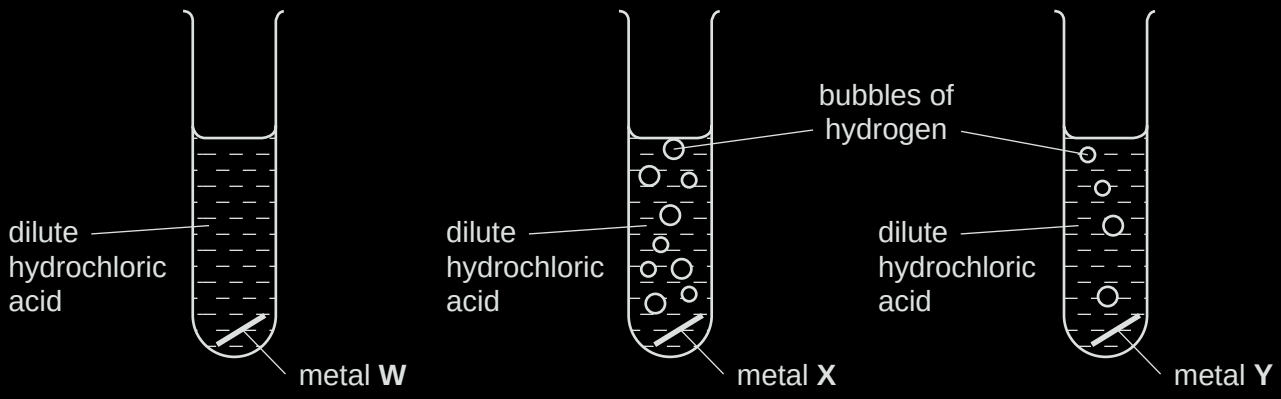
What is compound **X**?

A calcium hydroxide

B ethanoic acid

C sodium hydroxide

D sulphuric acid



What are metals **W**, **X** and **Y**?

	W	**X**	**Y**
A	copper	magnesium	zinc
B	copper	zinc	magnesium
C	magnesium	zinc	copper
D	zinc	magnesium	copper

30 Which statements about the pollutant carbon monoxide are correct?

1 It is a colourless, odourless gas.

2 It is formed by incomplete combustion of natural gas.

3 It reacts with haemoglobin in the blood.

A 1 and 2 only

B 1 and 3 only

C 2 and 3 only

D 1, 2 and 3

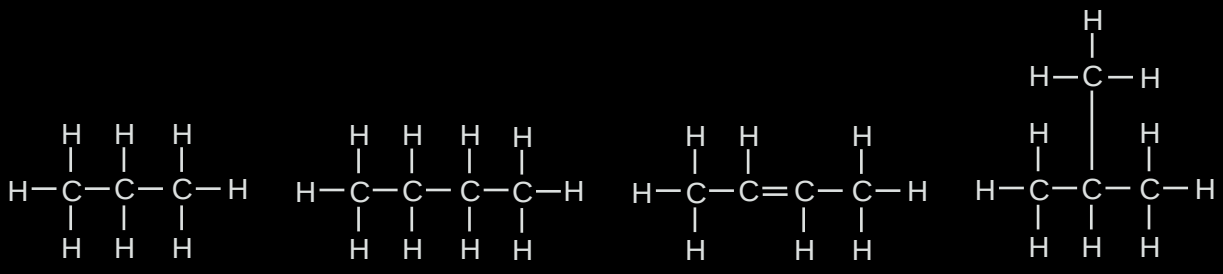
31 Which gas is **not** produced when hydrocarbons are burnt in the internal combustion engine?

A carbon dioxide

B carbon monoxide

C hydrogen

D nitrogen oxides



P Q R S

Which two hydrocarbons are isomers of each other?

A P and Q **B** P and S **C** Q and R **D** Q and S

34 When ethanol reacts with ethanoic acid, the ester ethyl ethanoate is formed.

$$C_2H_5OH + CH_3CO_2H \rightarrow CH_3CO_2C_2H_5 + H_2O$$

What is the formula of the ester formed when methanol reacts with butanoic acid ($C_3H_7CO_2H$)?

A $C_2H_5CO_2C_2H_5$

B $C_3H_7CO_2C_2H_5$

C $CH_3CO_2C_3H_7$

D $C_3H_7CO_2CH_3$

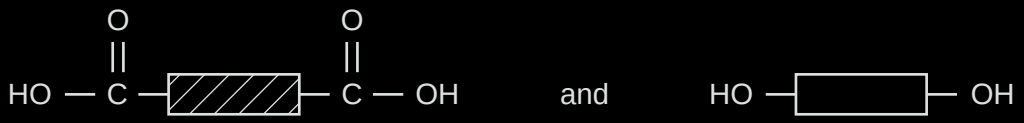
35 Which of these polymers is a protein?

A $(C_2H_3Cl)_n$

B $(C_2H_3NO)_n$

C $(C_5H_8O_2)_n$

D $(C_6H_{10}O_5)_n$



What is the macromolecule?

A a carbohydrate

B a polyamide

C a polyester

D a protein

DATA SHEET
The Periodic Table of the Elements

Group																	
I	II											III	IV	V	VI	VII	0
							1 H Hydrogen 1										4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulphur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	64 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
85 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57 *	178 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	Po Polonium 84	At Astatine 85	Rn Radon 86
Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

140 Ce Cerium 58	141 Pr Praseodymium 59	144 Nd Neodymium 60	Pm Promethium 61	150 Sm Samarium 62	152 Eu Europium 63	157 Gd Gadolinium 64	159 Tb Terbium 65	162 Dy Dysprosium 66	165 Ho Holmium 67	167 Er Erbium 68	169 Tm Thulium 69	173 Yb Ytterbium 70	175 Lu Lutetium 71
232 Th Thorium 90	Pa Protactinium 91	238 U Uranium 92	Np Neptunium 93	Pu Plutonium 94	Am Americium 95	Cm Curium 96	Bk Berkelium 97	Cf Californium 98	Es Einsteinium 99	Fm Fermium 100	Md Mendelevium 101	No Nobelium 102	Lr Lawrencium 103

*58-71 Lanthanoid series
90-103 Actinoid series

Key

a
X
b

a = relative atomic mass
X = atomic symbol
b = proton (atomic) number

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).